Department of Computer Science & Engineering

Syllabus for Placements Preparation

Data Structures and Algorithms

Data Structures: Arrays, Strings (Sliding Window, Two Pointers), Linked Lists (SLL, DLL, Circular), Stacks & Queues (Applications, Monotonic Stack, Trees (Binary Tree, BST, Traversals), Heaps (Min/Max Heap, Priority Queue), Hashing (HashMap, HashSet), Graphs (Adjacency Matrix/List)

Analysis & Design of Algorithms (ADA)

Complexity Analysis: Asymptotic Notations: O, Ω, Θ, Best, Average and Worst-Case Analysis, Recurrence Relations (Substitution, Master Theorem)

Algorithm Design Techniques: Brute Force, Divide and Conquer, Greedy Method, Dynamic Programming, Backtracking, Branch and Bound

Algorithms: Sorting: Bubble, Selection, Insertion, Merge, Quick, Heap. Searching: Linear, Binary (Variations), Recursion & Backtracking, Greedy Algorithms, Divide & Conquer, Dynamic Programming (Knapsack, LCS, LIS), Graph Algorithms (BFS, DFS, Dijkstra, Kruskal, Prim)

Classical Algorithms: Binary Search, Matrix Chain Multiplication, Floyd–Warshall Algorithm, Bellman–Ford Algorithm, Travelling Salesman Problem (TSP), N-Queen Problem

Computer Organization and Operating Systems (OS)

OS Fundamentals: OS Functions & Types, System Calls, Process vs Thread

Process Management: Process States & PCB, CPU Scheduling Algorithms (FCFS, SJF, Priority, RR), Context Switching

Synchronisation: Critical Section Problem, Semaphores, Mutex. Deadlock: Conditions, Prevention, Avoidance, Banker's Algorithm

Memory Management: Paging, Segmentation, Virtual Memory, Page Replacement Algorithms (FIFO, LRU, Optimal), Cache Memory, Cache Mapping Techniques

File & I/O Systems: File Allocation Methods, Disk Scheduling (FCFS, SSTF, SCAN, C-SCAN)

Computer Networks (CN)	
Networking Fundamentals: What is a computer network?, LAN vs MAN vs WAN, Bandwidth vs Throughput vs Latency, Hub vs Switch vs Router, Circuit switching vs Packet switching. **OSI & TCP/IP Models:** OSI 7 layers – functions of each layer, TCP/IP model and its layers, OSI vs TCP/IP comparison, Protocols at each layer, Why OSI is a reference model **Network Layer & IP Addressing:** IPv4 vs IPv6, Public vs Private IP address, Subnetting (basic interview numericals), CIDR, Routing: Distance Vector vs Link State, Router vs Switch. **Transport Layer:** TCP vs UDP (very frequently asked), TCP reliability mechanisms, Flow control vs Congestion control, Three-way handshake, TCP connection termination, Port numbers and well-known ports **Physical & Data Link Layer:** Encoding Techniques, MAC address, Error detection techniques (Parity, CRC), CSMA/CD vs CSMA/CA,Flow Control & Access Protocols, Stop-and-Wait vs Sliding Window. **Application Layer :**HTTP vs HTTPS, FTP vs SFTP, SMTP vs POP3 vs IMAP, DNS working, Cookies vs Sessions, What happens when you type a URL in a browser?	
Java Fundamentals	
Java Fundamentals: Java Features & JVM Architecture, JDK, JRE, JVM – Differences, Data Types, Variables, Type Casting. Operators & Expressions, Control Statements (if, switch, loops), Command Line Arguments **Object-Oriented Programming (OOP) in Java:** Classes & Objects, Constructors, this keyword, Inheritance, Method Overriding & Polymorphism, Abstraction (Abstract Classes & Interfaces), Encapsulation, final, static, super keywords, Interview Focus: Real-world OOP examples **Java Memory Management:** Stack vs Heap Memory, Object Creation & Lifecycle, Garbage Collection (GC), Types of GC, Memory Leaks & Prevention, **Exception Handling:** Types of Exceptions (Checked & Unchecked). try, catch, finally. throw vs throws, Custom Exceptions, Exception Handling Best Practices **Java Packages & Access Modifiers:** Packages (Built-in & User-Defined), Access Modifiers (public, private, protected, default), import keyword, Naming Conventions **Multithreading & Concurrency:** Thread Lifecycle, Creating Threads (Thread class, Runnable), Thread Synchronization, Inter-Thread Communication, Deadlock, Starvation, volatile, wait(), notify(). **Java Collections Framework:** List, Set, Map Interfaces. ArrayList vs LinkedList. HashSet vs TreeSet. HashMap vs TreeMap vs Hashtable. Iterators & Comparable vs Comparator. Time Complexity of Collections.	

Strings & Wrapper Classes: String, StringBuffer, StringBuilder. String Pool & Immutability. Wrapper Classes, Autoboxing & Unboxing

for **Placement Coordinators**

Signature of HOD